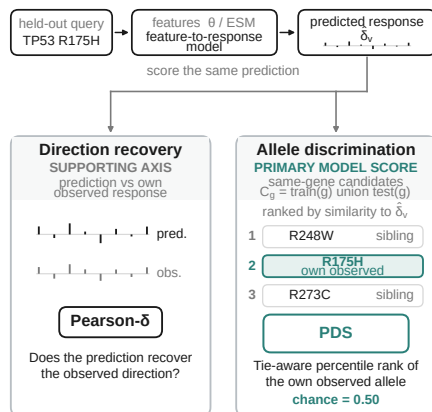
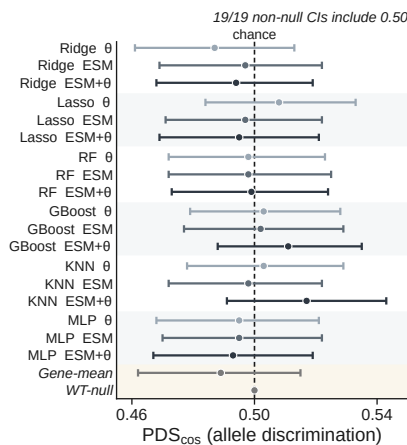


a

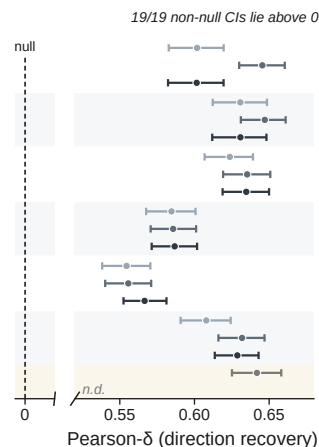


Held-out queries only; candidate set contains all same-gene variants.

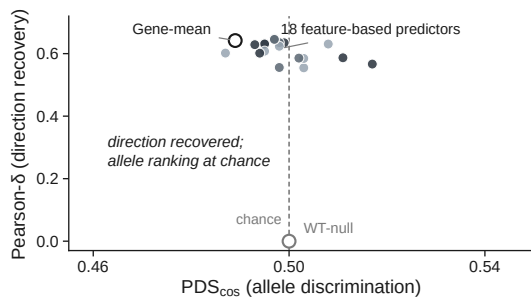
b

• θ • ESM • ESM+ θ • reference

c



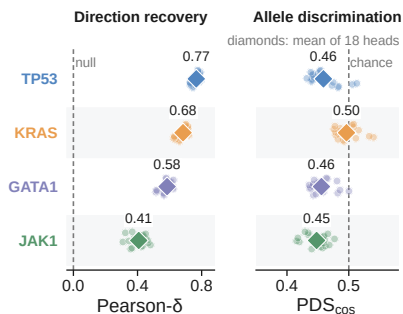
d



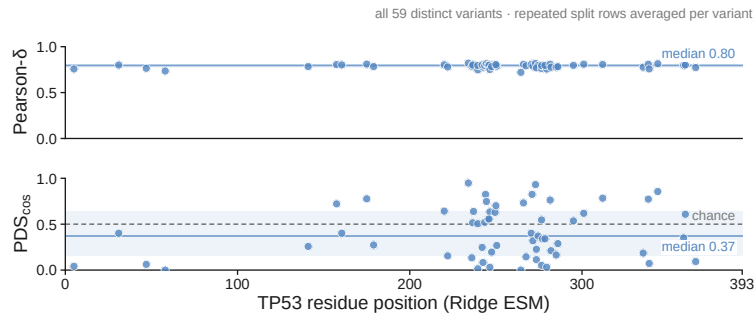
e



f



g



h

